
Introduction to HTML

Learning Outcomes

See how **HTML** describes the structure of a web page

Learn how tags or elements are added to your document

Explore the HTML elements specification

HTML

HTML stands for **HyperText Markup Language**

Newspaper Page components

Headlines,
Subheadings,
Paragraphs,
Text,
Image,



HTML describes structure of pages

```
<html>

<body>
  <h1>This is the Main Heading</h1>
  <p>This text might be an introduction to the rest of the page. And if the page is a long one it might
    be split up into several sub-headings.</p>
  <h2>This is a Subheading</h2>
  <p>Many long articles have subheadings to help you follow the structure of what is being written.
    There may even be sub-sub-headings (or lower-level headings).</p>
  <h2>Another Subheading</h2>
  <p>Here you can see another subheading.</p>
</body>

</html>
```

HTML

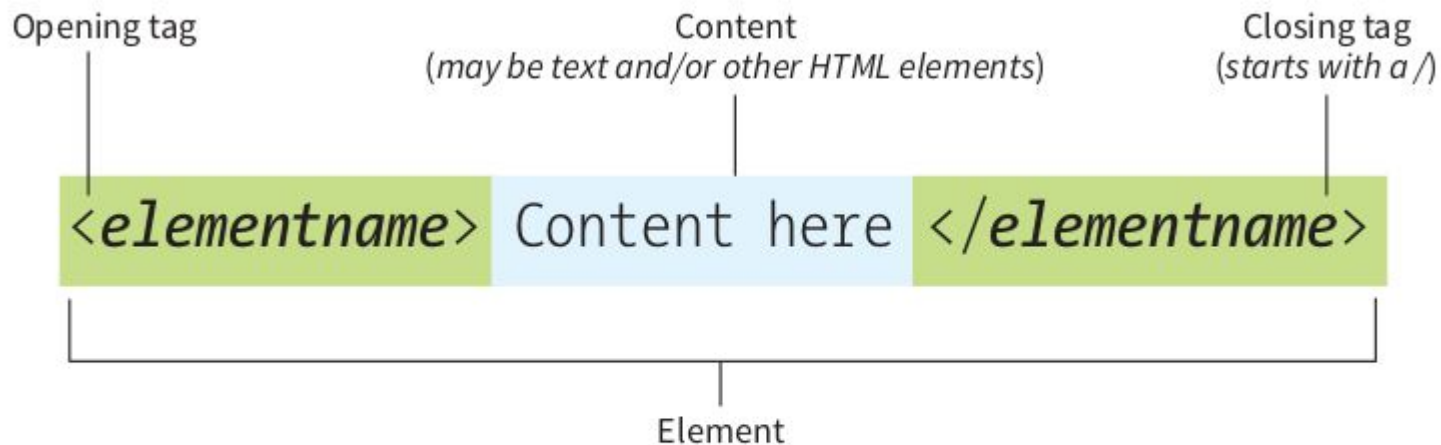
HTML code is made up of characters that live inside **angled brackets** called **HTML elements**

HTML Elements are usually made up of two **tags**:

- an opening tag and a closing tag

Each HTML element tells the browser something about the information that sits between its opening and closing tags

HTML Tag



Example:

```
<h1>Black Goose Bistro</h1>
```

HTML Tag Attributes

Attributes provide additional information about the contents of an element

They appear on the opening tag of the element and are made up of two parts: a name and a value, separated by an equals sign



The diagram illustrates the structure of an HTML tag attribute. It shows the opening tag `<p lang="en-us">` with the text "Paragraph in English" and the closing tag `</p>`. The attribute `lang="en-us"` is highlighted. A bracket above the attribute points to the text "ATTRIBUTE NAME", and a bracket below it points to the text "ATTRIBUTE VALUE".

```
graph TD
    AN[ATTRIBUTE NAME] --- A["<p lang="en-us">Paragraph in English</p>"]
    AV[ATTRIBUTE VALUE] --- A
```


Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

DOCTYPE identifies the document as written in **HTML5**

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

The **html** element is called the **root element** because it contains all the elements in the document

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

The **head** element contains elements that pertain to the document that are not rendered as part of the content, such as its **title**, **style sheets**, **scripts**, and **metadata**

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

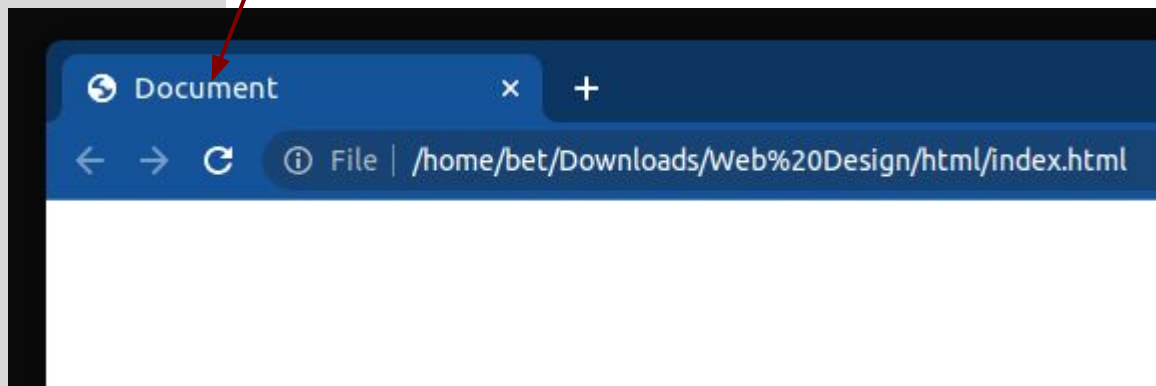
meta elements provide document metadata, information about the document

Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

title mandatory element



Basic HTML document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Document</title>
  </head>
  <body>

  </body>
</html>
```

body element contains everything that we want to show up in the browser window

Semantics

Elements, attributes, and attribute values in HTML are defined to have certain meanings (semantics)

Example:

the **ol** element represents an **ordered list**

the **lang** attribute represents the language of the content

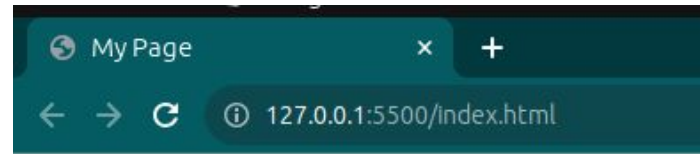
Semantics

Allows **HTML processors**, such as **web browsers** or **search engines**, to present and use documents and applications in a wide variety of contexts

Consider as an example the html page shown on the right

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Welcome to my page</h1>
    <p>I love adventure!</p>
    <h2>Where I live</h2>
    <p>I live in Ethiopia!</p>
  </body>
</html>
```

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Welcome to my page</h1>
    <p>I love adventure!</p>
    <h2>Where I live</h2>
    <p>I live in Ethiopia!</p>
  </body>
</html>
```



Welcome to my page

I love adventure!

Where I live

I live in Ethiopia!

Browsers might show headings in large letters and bold

Semantics

Speech browser might use a different volume or a slower voice to read headings shown on the right

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Welcome to my page</h1>
    <p>I love adventure!</p>
    <h2>Where I live</h2>
    <p>I live in Ethiopia!</p>
  </body>
</html>
```

Semantics

Search engines can use the headings to more **effectively index a page**, or to provide quick links to subsections of the page from their results.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Welcome to my page</h1>
    <p>I love adventure!</p>
    <h2>Where I live</h2>
    <p>I live in Ethiopia!</p>
  </body>
</html>
```

Semantics

Different **tools** can use the headings to create a table of contents

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Welcome to my page</h1>
    <p>I love adventure!</p>
    <h2>Where I live</h2>
    <p>I live in Ethiopia!</p>
  </body>
</html>
```

Semantics

The following snippet is intended to represent the heading of a corporate site.

Is it correct?

```
<body>  
  <h1>ACME Corporation</h1>  
  <h2>The leaders in arbitrary fast delivery since 1920</h2>  
  ...  
</body>
```

Semantics

It is **non-conforming** because the **second line is not intended to be a heading of a subsection**, but merely a

subheading or subtitle (a subordinate heading for the same section)

```
<body>
  <h1>ACME Corporation</h1>
  <h2>The leaders in arbitrary fast delivery since 1920</h2>
  ...
</body>
```

Semantics

The **hgroup** element can be used for showing heading and subheadings

```
<body>
  <hgroup>
    <h1>ACME Corporation</h1>
    <p>The leaders in arbitrary fast delivery since 1920</p>
  </hgroup>
  ...
</body>
```


Semantics

What is the problem with the html shown below

```
<body>
  <table>
    <tr>
      <td>The only journey is the one within.</td>
    </tr>
    <tr>
      <td>
        –<a href="https://quotes.org/~maria-rilke/">
          <cite>Rainer Maria Rilke</cite></a>
          >, 1915
        </td>
      </tr>
    </table>
  </body>
```

Semantics

The correct one

```
<body>
  <blockquote>
    <p>The only journey is the one within.</p>
  </blockquote>
  <p>
    -<a href="https://quotes.org/~maria-rilke/">
      <cite>Rainer Maria Rilke</cite></a>
    >, 1915
  </p>
</body>
```

Semantic HTML Elements

<https://html.spec.whatwg.org/>

https://www.w3schools.com/html/html5_semantic_elements.asp

Semantic Elements in HTML5

<https://html.spec.whatwg.org/#semantics>

Semantic Elements in HTML5

<https://html.spec.whatwg.org/multipage/sections.html#sections>

Semantic Elements

`<article>`

`<aside>`

`<details>`

`<figcaption>`

`<figure>`

`<footer>`

`<header>`

`<main>`

`<mark>`

`<nav>`

`<section>`

`<summary>`

`<time>`



<header>

<nav>

<header>

<article>

<header>

<section>

<section>

<footer>

<aside>

<section>

<section>

<section>

<footer>

<main>

The **main** content area of a document **includes content that is unique to that document** and **excludes content that is repeated across a set of documents** such as

site navigation links, copyright information, site logos and banners, and search forms (unless the document or application's main function is that of a search form)

<main>

A document **must not have more than one main element** that does not have the hidden attribute specified

<https://html.spec.whatwg.org/#the-main-element>

<section>

The **section** element is used to define a generic section of a document or application

A section, in this context, is a **thematic grouping of content**, typically with a heading

<https://html.spec.whatwg.org/#the-section-element>



<section>

Examples of sections would be

chapters, the various tabbed pages in a tabbed dialog box, or the numbered sections of a thesis

A web site's home page could be split into **sections** for an **introduction**, **news items**, and **contact information**

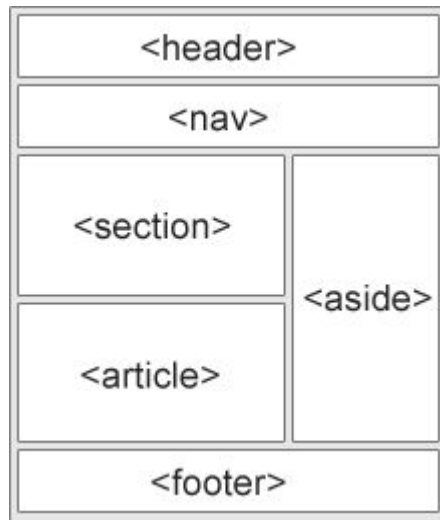
It's important to note that this element isn't intended for styling purposes



<nav>

The **nav** element is used to **wrap major navigational links** to other pages or parts within the same page

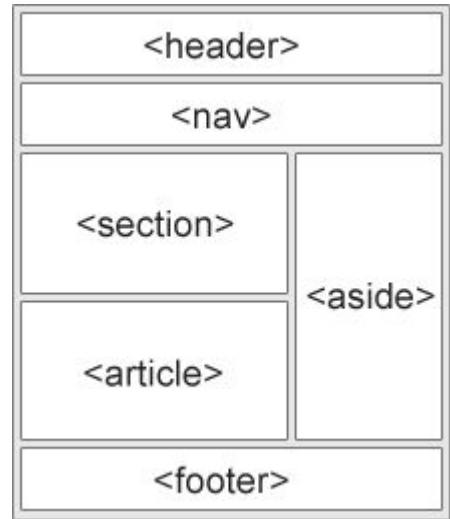
<https://html.spec.whatwg.org/multipage/sections.html#the-nav-element>



<article>

The **article** element **represents** a **complete**, or **self-contained**, composition in a document, page, application, or site and that is, in principle, **independently distributable** or **reusable**

<https://html.spec.whatwg.org/multipage/sections.html#the-article-element>

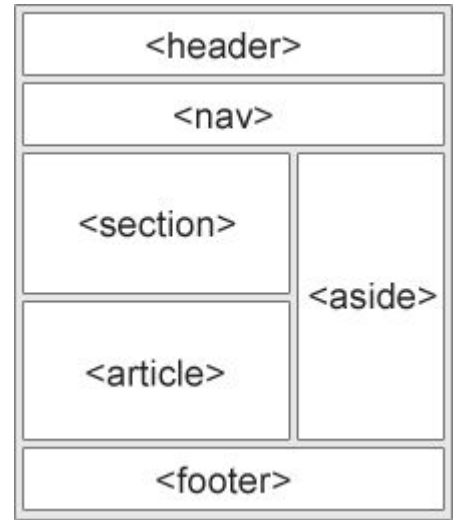


<article>

When **article** elements are **nested**, the inner article elements represent articles that are in principle **related to the contents of the outer article**

For instance,

a blog entry on a site that accepts user-submitted comments could represent the comments as article elements nested within the article element for the blog entry



<article>

Examples:

a **forum post**, a magazine or newspaper **article**, a **blog entry**, a **user-submitted comment**, an **interactive widget** or **gadget**, or any other independent item of content



<aside>

The **aside** element is used for content that is tangentially related to the content around it

It can be **used for side-bars, pull quotes, advertising, and groups of navigation elements**

<https://html.spec.whatwg.org/multipage/sections.html#the-aside-element>



Example pull quote

Drawing conclusions is our job, and is based on contextual, relevant concerns that we discover and define in our design process.

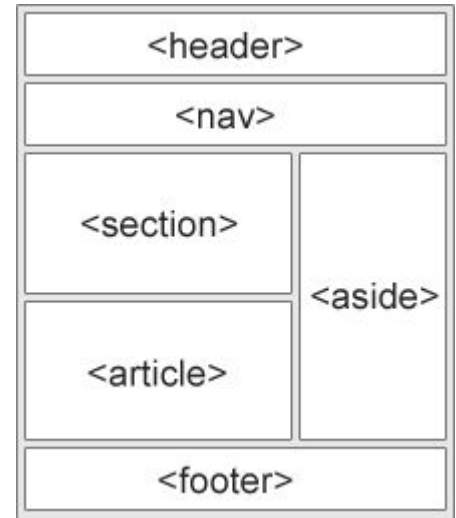
accessible and well designed—so long as the rest of the

As with any o
you say on a
it, and how yo
clearly, it is d
have 50,000
as the actiona

<header>

The **header** element represents a **group of introductory or navigational aids**

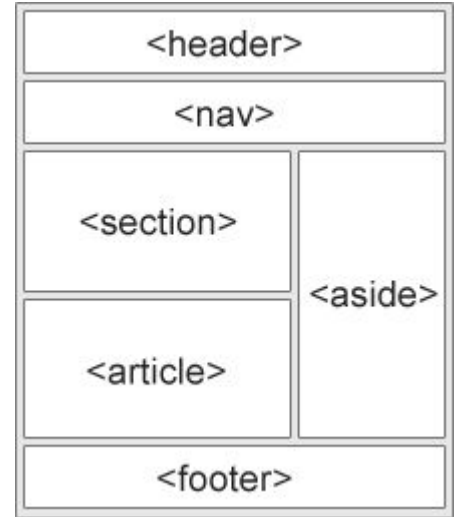
A **header** element is intended to usually contain a heading (an **h1–h6** element or an **hgroup** element), but this is not required



<header>

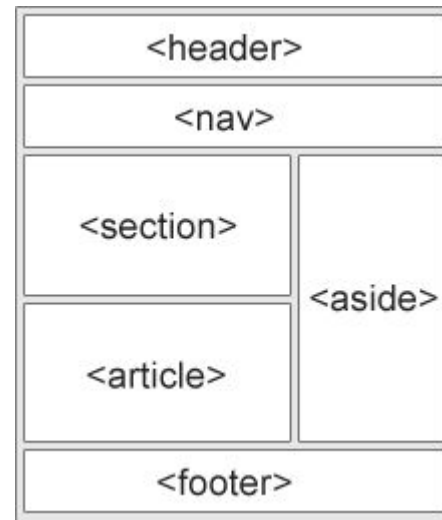
The header element can also be used to wrap a **section's table of contents**, a **search form**, or **any relevant logos**

<https://html.spec.whatwg.org/multipage/sections.html#the-header-element>



<footer>

The **footer** element represents **a footer for its nearest ancestor sectioning** content element, or **for the body element if there is no such ancestor**



<footer>

A **footer** typically contains information about its section such as

**who wrote it, links to related documents,
copyright data, and the like**

<https://html.spec.whatwg.org/multipage/sections.html#the-footer-element>



<h1>,<h2>,<h3>,<h4>,<h5>,<h6>

These elements represent headings for their sections

<https://html.spec.whatwg.org/multipage/sections.html#the-h1,-h2,-h3,-h4,-h5,-and-h6-elements>

<hgroup>

The **hgroup** element represents a heading and related content

The element may be used to group an **h1–h6** element with one or more **p** elements containing content representing a **subheading, alternative title, or tagline**

<https://html.spec.whatwg.org/multipage/sections.html#the-hgroup-element>

<hgroup>

```
<hgroup>
```

```
  <h1>The reality dysfunction</h1>
```

```
  <p>Space is not the only void</p>
```

```
</hgroup>
```

Grouping Contents

<https://html.spec.whatwg.org/multipage/grouping-content.html>

<P>

The **p** element represents a paragraph

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-p-element>

<hr>

The **hr** element represents a paragraph-level thematic break, e.g. a scene change in a story, or a transition to another topic within a section of a reference book

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-hr-element>

<blockquote>

The **blockquote** element represents a section that is quoted from another source

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-blockquote-element>

The `ol` element represents **a list of items**, where the **items have been intentionally ordered**, such that **changing the order would change the meaning of the document**

`https://html.spec.whatwg.org/multipage/grouping-content.html#the-ol-element`

The `ul` element represents **a list of items**, where the **order of the items is not important** — that is, where changing the order would not materially change the meaning of the document

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-ul-element>

<menu>

The **menu** element represents a **toolbar** consisting of its contents, in the form of an unordered list of items (represented by **li** elements), each of which represents a command that the user can perform or activate

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-menu-element>

<figure>

The **figure** element represents some flow content, optionally with a caption, that is self-contained (like a complete sentence) and is typically referenced as a single unit from the main flow of the document

The element can be used to **annotate illustrations, diagrams, photos, code listings**, etc.

<https://html.spec.whatwg.org/multipage/grouping-content.html#the-menu-element>

<div>

The **div** element has no special meaning at all

It represents its children

It can be used with the **class**, **lang**, and **title** attributes to mark up semantics common to a group of consecutive elements.

Text-Level Semantics

`https://html.spec.whatwg.org/multipage/text-level-semantics.html#usage-summary`

<a>

If the **a** element has an **href** attribute, then **it represents a hyperlink** (a hypertext anchor) labeled by its contents

The **target**, **download**, **ping**, **rel**, **hreflang**, **type**, and **referrerpolicy** attributes **must be omitted if the href attribute is not present**

```
https://html.spec.whatwg.org/multipage/text-level-semantics.  
html#the-a-element
```


The **em** element represents **stress emphasis** of its contents

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-em-element>

The **strong** element represents **strong importance, seriousness, or urgency** for its contents

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-strong-element>

<data>

The **data** element represents its contents, along with a machine-readable form of those contents in the value attribute

The value attribute must be present. Its value must be a representation of the element's contents in a machine-readable format

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-data-element>

<sup>, <sub>

The **sup** element represents a superscript and the **sub** element represents a subscript

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-sub-and-sup-elements>



The **i** element represents a span of text in an **alternate voice** or **mood**, or otherwise **offset from the normal prose** in a manner indicating a **different quality of text**, such as a **taxonomic designation**, a **technical term**, an idiomatic phrase from another language, transliteration, a thought

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-i-element>



The **b** element represents a span of text to which **attention is being drawn** for utilitarian purposes without conveying any extra importance and with no implication of an alternate voice or mood, such as

keywords in a document abstract, product names in a review, actionable words in interactive text-driven software, or an article lede

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-b-element>

The **span** element doesn't mean anything on its own, but can be useful when used together with the global attributes, e.g. **class**, **lang**, or **dir**

It represents its children

The **br** element represents a line break.

br elements must be used only for line breaks that are actually part of the content, as in poems or addresses.

<https://html.spec.whatwg.org/multipage/text-level-semantics.html#the-br-element>

Embedded content

<https://html.spec.whatwg.org/multipage/embedded-content.html>

<picture>

The **picture** element is a container which **provides multiple sources to its contained `img` element to allow authors to declaratively control or give hints to the user agent about which image resource to use**, based on the **screen pixel density, viewport size, image format**, and other factors

It represents its children

<https://html.spec.whatwg.org/multipage/embedded-content.html#the-picture-element>

An **img** element represents an image

The image given by the **src** and **srcset** attributes is the embedded content

The value of the **alt** attribute provides equivalent content for those who cannot process images or who have image loading disabled

<https://html.spec.whatwg.org/multipage/embedded-content.html#the-img-element>

<https://html.spec.whatwg.org/multipage/images.html>

`<img>`

The `src` attribute must be present, and must contain a valid non-empty `URL`

<iframe>

The **iframe** element represents its nested navigable

The **src** attribute gives the URL of a page that the element's nested navigable is to contain. The attribute, if present, must be a valid non-empty URL

<https://html.spec.whatwg.org/multipage/iframe-embed-object.html#the-iframe-element>

<video>

A **video** element is used for playing videos or movies, and audio files with captions

Has **src**, **crossorigin**, **preload**, **autoplay**, **loop**, **muted**, and **controls** attributes

The **poster** attribute gives the URL of an image file that the user agent can show while no video data is available

<https://html.spec.whatwg.org/multipage/media.html#the-video-element>

<audio>

An **audio** element represents a sound or audio stream

Has **src**, **crossorigin**, **preload**, **autoplay**, **loop**, **muted**, and **controls** attributes

<https://html.spec.whatwg.org/multipage/media.html#the-audio-element>

Forms

<https://html.spec.whatwg.org/multipage/forms.html#forms>

<form>

The **form** element represents a hyperlink that can be manipulated through a collection of form-associated elements, some of which can represent editable values that can be submitted to a server for processing

<https://html.spec.whatwg.org/multipage/forms.html#forms>

Tabular Data

<https://html.spec.whatwg.org/multipage/tables.html#tables>

<table>

The **table** element represents data with more than one dimension, in the form of a table

Tables have **rows**, **columns**, and **cells** given by their descendants

The rows and columns form a **grid**; a table's cells must completely cover that grid without overlap

<https://html.spec.whatwg.org/multipage/tables.html#tables>

What HTML tag/element
do you use for which part?
Why?



Additional Resources

<https://webflow.com/blog/html5-semantic-elements-and-webflow-the-essential-guide>

Additional Resources

<https://developer.mozilla.org/en-US/docs/Learn/HTML>

https://www.w3schools.com/tags/ref_byfunc.asp

https://www.washington.edu/accesscomputing/webd2/student/unit2/common_tags.html

<https://htmlreference.io/>